

RUSSELL ELLIOTT

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EDUCATION

University of California Santa Cruz, Santa Cruz, CA

Master of Science in Computer Science

September 2024 - December 2025

Bachelor of Science in Computer Science

June 2020 - June 2024

Coursework: Full Stack Web Development, NLP, Deep Learning, Machine Learning, GenAI, Network Security, Computer Security, Distributed Systems, Mobile App Development, Database Systems, Computer Networks, Computer Systems & Design

SKILLS

Languages	Typescript, Javascript, SQL, Python, PHP, Swift, HTML, CSS, C, C++, Java
AI/ML Tools	TensorFlow, PyTorch, MLX, Hugging Face, LangChain, Numpy, Pandas, OpenAI
Web Development	ReactJS, AngularJS, Redux, NextJS, FastAPI, Flask, RESTful/GraphQL APIs, OpenAPI
Databases	PostgreSQL, MySQL, Firebase, MongoDB, LanceDB, Pinecone
Environments	AWS (EC2/S3), GCP, VirtualBox
Platforms	Docker, Github, Gitlab, Websockets, Cypress, Jest, React Testing

EXPERIENCE

AI Model Optimization Researcher

May 2025 - December 2025

iNRG Research Lab @ UCSC, Santa Cruz, CA

- Designed LLM-based indoor localization system using frozen LLaMA-3.2-1B with LoRA, achieving 2.84m MAE, surpassing traditional baselines.
- Enhanced prediction stability by replacing model head with lightweight MLP regressor for continuous coordinate outputs.
- Built WiFi telemetry preprocessing pipeline enabling sub-meter accuracy (0.88–0.96m MAE) with only 3–10% calibration data and sustaining 2.5–3.5m MAE under AP dropout, reducing calibration needs by 10x.

Full Stack Web Developer, Team Lead

January 2023 - June 2024

Tech4Good @ UCSC, Santa Cruz, CA

- Automated data pipelines and integrated APIs (e.g., TikTok) to power research and career platforms with real-world insights, using AngularJS, Firebase, and JavaScript.
- Improved user engagement by building text-annotation UI components and career video displays in AngularJS.
- Co-managed 100+ member lab, organizing tasks, prioritizing web app projects, and tracking progress in Airtable.

Web Developer, AI Integration Specialist

July 2022 - January 2024

sierrahiking.net, Remote

- Improved user navigation and engagement by designing and developing hiking-focused WordPress site with interactive master map and detailed trail pages.
- Enhanced trip-planning user experience by building WordPress plugin with OpenAI to generate custom backpacking lists.

LEADERSHIP

- UCSC Blueprint:** Mobile Developer. Developed app to log volunteer hours for endangered/native animals.
- UCSC ACM:** Event Coordinator. Hosted hackathons and other events to promote collaborative software development.
- UCSC CruzHacks:** Software Engineer. Developed hackathon website/application and judging portal for projects.
- UCSC Chess Club:** President. Hosted on-campus events and tournaments as well as events with other clubs.

PROJECTS

POI Validation Pipeline. Built 3-stage pipeline cleaning restaurant data across 300+ chains for Overture Maps using deterministic Google Places status detection and hybrid LLM data retrieval and augmentation with traceable sources. ([GitHub](#))

Apple Notes Semantic Search. Building MCP-powered note management system with semantic search via local LLMs, BERTopic clustering visualization, RAG pipelines, and LanceDB vector database over Apple Notes source data. ([GitHub](#))

LLM Security Benchmarking. Evaluated 5 LLMs across 25 threat categories, proving jailbreak prompts bypass safety filters (92% success rate). Built pipeline tracking rejection behaviors and cosine similarity of generated exploits. ([Scripts](#)) ([Graphs](#))